

# Mechanistic simulation of heritable expression states, cell division and drug response in single-cell populations

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## Abstract

Heritable differences in gene expression decide which cancer cells survive a drug, yet the simulators used to benchmark single-cell methods model neither growth nor division, and cannot generate such states. Here we present Biomodelling.jl 2.0, in which stochastic reaction kinetics run inside growing, dividing and drug-treated cells. Exact, hybrid and adaptive kernels are coupled to exponential growth, volume-scaled transcription, gene replication and division with binomial partitioning, to dose schedules and state-dependent killing, and to a lineage record, observation models and likelihood-free inference. Expression memory, cell-size scaling and cell-cycle-dependent bursting emerge from this physiology rather than being imposed. A single resistance gene with slow promoter switching reproduces the lineage and memory signatures of heritable drug tolerance, and the framework supplies ground truth that benchmarks of memory genes, network inference and perturbation prediction lack. Calibrated to time-lapse measurements of cisplatin-treated cells, it reproduces lineage correlations and the dose-invariance of single-cell timing, and its residual errors expose the limits of one switching gene. With a built-in optimiser it reproduces the opposite rankings of intermittent dosing reported in a melanoma trial and in xenografts, and shows the ranking to turn on two measurable cell properties, their growth under drug and their fitness without it.

Isogenic cells differ in their transcriptional states, those differences are often inherited across several divisions, and the inherited state biases what a cell does under stress. Cancer cells that survive targeted or cytotoxic therapy frequently do so without a resistance mutation: they occupy a rare, reversible, drug-tolerant expression state that pre-exists the treatment or is induced by it<sup>1–5</sup>. Lineage-resolved experiments now measure this memory directly. The fates of sisters and cousins are correlated, clonal barcode diversity is preserved or collapsed depending on whether tolerance pre-exists or is induced, and the timescale of expression fluctuations relative to the cell cycle decides which genes carry memory at all<sup>6–9</sup>. In parallel, mechanistic analyses of single-cell RNA-sequencing data have shown that transcriptional kinetics can be inferred correctly only when gene copy number, cell age and partitioning at division are modelled explicitly<sup>10–12</sup>, and evaluations of “virtual cell” models for perturbation prediction keep finding that deep-learning approaches fail to beat simple baselines, largely for want of mechanism and causal structure<sup>13–16</sup>.

These developments create a demand that existing simulators do not meet. The simulators used to benchmark single-cell methods, among them SERGIO, dyngen, scMultiSim and GRouNdGAN<sup>17–20</sup>, model regulation and technical noise but not cell growth, volume-dependent transcription, gene replication or the partitioning of molecules at division; lineage simulators such as TedSim<sup>21</sup> generate division trees but attach phenomenological, largely Markovian expression to them. None of them can therefore produce expression memory, cell-size scaling, cell-cycle-dependent bursting or persister dynamics from first principles, and none provides the ground truth that the methods now being developed for exactly those phenomena require<sup>22</sup>. Stochastic simulators of growing and

dividing cells exist<sup>23,24</sup> but stop at the single-cell state: they offer no drug response, no lineage statistics, no observation models, no inference and no benchmark generation.

Here we present Biomodelling.jl 2.0, a rewrite of a Julia framework whose distinguishing feature, the coupling of stochastic reaction kinetics to exponential growth, volume-scaled transcription and binomial partitioning at division<sup>25</sup>, is precisely the mechanism behind heritable expression states. Version 2.0 turns that feature into a complete framework for the mechanistic study of drug tolerance and for single-cell benchmarking. We validate the engine against exact stationary laws and an independent implementation, show that growth, division and partitioning alone reproduce published results on cell-size scaling, partitioning noise, lineage-versus-population statistics and cell-cycle-dependent bursting, and demonstrate that a single resistance gene with slow promoter switching reproduces the quantitative lineage and memory signatures of heritable drug tolerance. We then use the framework as a ground-truth generator for memory-gene detection, network inference, imputation and perturbation prediction; as an inference engine that recovers kinetic parameters from snapshot and lineage data, quantifies the bias of ignoring division, and maps the ridge to which drug parameters are determined by population data; as a model calibrated to published time-lapse measurements and confronted with conditions it was not fitted to; and, with a built-in schedule optimiser, as a testbed in which clinical dosing regimens are compared and better ones are sought.

## Results

### Layers couple reaction kinetics to growth and division

Biomodelling.jl 2.0 is organised in layers (Fig. 1). A model is a list of reactions with mass-action, Hill or custom kinetics and named parameters; promoter states are explicit species, grouped so that they are inherited at division and doubled at replication. Four kernels advance a cell's state at fixed volume: the exact direct method, fixed-step tau-leaping, a hybrid scheme that falls back to the exact method whenever a leap would produce a negative count, and adaptive tau-leaping with critical-reaction handling. The population layer grows each cell's volume exponentially, scales propensities with volume so that concentrations are invariant, replicates genes at a set point of the cycle, divides cells by a sizer, adder or timer rule with binomial or over-dispersed partitioning, and controls the population size by Moran-like replacement, free growth or a logistic hazard. A perturbation layer applies dose schedules, state-dependent death, growth inhibition, drug-induced rate changes and genetic perturbations. Every cell's ancestry, fate and state at birth and division enter a lineage table, from which heritability, autocorrelation, memory timescales, fluctuation-test and clonal-memory statistics follow. Observation models convert true counts into scRNA-seq, smFISH or time-lapse measurements, and the same forward model drives likelihood-free inference. Table 1 sets these capabilities against those of the simulators most used to benchmark single-cell methods.

Table 1. Capabilities of single-cell simulators (✓ native; ○ partial or phenomenological; – absent). SERGIO<sup>17</sup>, dyngen<sup>18</sup>, scMultiSim<sup>19</sup>, GRouNdGAN<sup>20</sup>, TedSim<sup>21</sup>, Cassiopeia<sup>26</sup>, ABM.jl (AgentBasedModeling.jl)<sup>24</sup>, Biomodelling.jl v1.x<sup>25</sup> and v2.0 (this work). Cassiopeia simulates division topologies with heritable fitness and CRISPR barcodes rather than molecular content, and its inference recovers phylogenies rather than kinetic parameters; entries were read from each tool’s publication and documentation.

| Capability                                           | SERGIO | dyngen | scMulti-Sim | GRouNd-GAN | TedSim | Cassio-peia | ABM.jl | v1.x | v2.0 |
|------------------------------------------------------|--------|--------|-------------|------------|--------|-------------|--------|------|------|
| Mechanistic reaction kinetics (SSA / tau-leaping)    | ○      | ✓      | ○           | –          | –      | –           | ✓      | ✓    | ✓    |
| Gene regulatory networks with known ground truth     | ✓      | ✓      | ✓           | ✓          | –      | –           | ○      | ✓    | ✓    |
| Cell growth, volume-scaled transcription             | –      | –      | –           | –          | –      | –           | ✓      | ✓    | ✓    |
| Gene replication / cell- cycle copy number           | –      | –      | –           | –          | –      | –           | –      | –    | ✓    |
| Division with molecular partitioning                 | –      | –      | –           | –          | ○      | ○           | ✓      | ✓    | ✓    |
| Heritable promoter states, expression memory         | –      | –      | –           | –          | ○      | –           | ○      | ○    | ✓    |
| Lineage tree with per-cell fates                     | –      | –      | –           | –          | ✓      | ✓           | ○      | –    | ✓    |
| Drug response: dose schedules, state-dependent death | –      | –      | –           | –          | –      | –           | –      | –    | ✓    |
| Genetic perturbations (knockdown / overexpression)   | ✓      | ✓      | ○           | ✓          | –      | –           | ○      | –    | ✓    |
| scRNA-seq observation model                          | ✓      | ✓      | ✓           | ✓          | ✓      | –           | –      | ○    | ✓    |
| smFISH / time-lapse observation                      | –      | –      | –           | –          | –      | –           | –      | –    | ✓    |
| Parameter inference from data                        | –      | –      | –           | –          | –      | ○           | –      | –    | ✓    |
| AnnData output                                       | ✓      | ✓      | ✓           | ✓          | ○      | ○           | –      | –    | ✓    |

## The engine matches exact laws and an independent package

We first verified the kernels against exact stationary laws (Fig. 2). For a birth-death process the simulated distribution of 50,000 cells is indistinguishable from the Poisson law (Fig. 2a); for the telegraph model in an intermediate and in a bursty regime it matches the Beta-Poisson distribution<sup>27</sup> (Fig. 2b); and for bursty protein production it matches the negative binomial (Fig. 2c). Across kernels, Kolmogorov-Smirnov distances to the exact law with 20,000 cells lie below the 99% critical value for the direct method, for the adaptive scheme at both tolerances and for the hybrid scheme at the shorter step, and above it for the hybrid scheme at the longer step (0.0160 and 0.0122 against a critical value of 0.0115; Fig. 2d). That is the expected behaviour of an approximate kernel, and it sets the step below which the approximation is not detectable at this sample size; wall-clock costs are 0.25 s to 0.91 s per  $10^4$  cells. Simulating the same telegraph model with JumpProcesses.jl gives distributions whose two-sample Kolmogorov-Smirnov statistic against Biomodelling.jl is 0.0034 (99% critical value 0.0163; Fig. 2e), with a 6.2-fold (in favour of JumpProcesses.jl) difference in serial wall-clock time. Population simulations scale linearly with the number of cells and approximately linearly with the number of reactions: 2.6 s for 5000 cells  $\times$  10 telegraph genes  $\times$  200 steps (hybrid kernel) with four threads (Fig. 2f and Supplementary Table 3).

## Physiology generates memory, scaling and cycle effects

The physiology of the population layer suffices to produce properties of single-cell data that phenomenological simulators must impose by hand (Fig. 3). In a single lineage of a slowly switching gene, mRNA and protein track

volume within a cycle and are halved at division, while the promoter state, inherited by both daughters, persists over many divisions (Fig. 3a). For a fast-switching gene without replication, transcript numbers scale linearly with cell volume, so that concentration is homeostatic; adding mid-cycle gene replication doubles transcription in the second half of the cycle and lifts counts of large cells above the volume-scaling line (Fig. 3b), the copy-number effect that mechanistic fits to scRNA-seq data must account for<sup>10</sup>. Correlations of protein concentration at division between mothers and daughters, sisters and cousins fall from 0.94 to 0.02 as the promoter switching rate crosses the inverse cell-cycle time (Fig. 3c), mother-daughter and sister pairs correlating similarly at slow switching and mother-daughter pairs more strongly as switching quickens, cousins least, and mRNA less heritable than protein. The memory timescale extracted from the generation-resolved lineage autocorrelation is set by the promoter switching time and is nearly independent of the cell-cycle time as long as memory outlasts the cycle; when the two are comparable the estimate is bounded below by one cell cycle, the resolution of the autocorrelation (Fig. 3d). As reported for real lineages<sup>11</sup>, the coefficient of variation of a stable protein's concentration is larger across the population than along single lineages, whether or not cell-cycle times are noisy, and over-dispersed partitioning adds variability in both perspectives (Fig. 3e). Telegraph parameters fitted separately to cells before and after replication differ by the expected factor of two in burst frequency, and a fit that ignores the cell cycle returns intermediate values (Fig. 3f).

## One resistance gene reproduces tolerance signatures

To test whether heritable expression states suffice to reproduce the lineage and memory signatures of drug-tolerant persistence, we coupled a resistance gene with slow promoter switching (memory timescale 45 time units, cell-cycle time 20) to a death hazard that rises with dose and is suppressed by the concentration of the gene's protein product (Methods). Under continuous, pulsed and reduced-dose schedules the population first collapses as low-expressing cells die, then regrows from high-expressing survivors (Fig. 4a); kill curves are biphasic with a surviving fraction that falls with dose (Fig. 4b). Between doses 0.5 and 2 the net rate of population decline over the first 30 time units of treatment rises from 0.023 to 0.050 per time unit (2.1-fold), while the mean division time of cells born under drug changes by 0.0% and the mean time to death of killed cells moves only from 8.5 to 6.2 time units (Fig. 4c; the single-cell timing distributions are in Supplementary Fig. 1). This reproduces the observation that population decay depends strongly on dose even though single-cell division and death times barely do<sup>6</sup>, because fate is biased by states that exist before the drug arrives. Unlike the growth-arrested persisters of the canonical definition<sup>1</sup>, the survivors here divide at an essentially unchanged rate (Fig. 4c, inset), so the model reproduces the heritable-state selection signatures of tolerance rather than its proliferative-arrest hallmark.

Related cells accordingly share the fate of their lineages. Among cells born in the 25 time units before the drug, one whose sister founded a surviving lineage does so itself with probability 0.39, against a marginal probability of 0.08 (4.8-fold enrichment over 1830 sister pairs; 2.9-fold for cousins), whereas in a fast-switching control, in which survival is rarer, the conditional and marginal probabilities coincide (0.03 versus 0.02; Fig. 4d). When tolerance pre-exists, survivors descend from a minority of founder clones and the effective number of clones drops from 491 to 89; when tolerance is drug-induced, so that the drug raises the promoter activation rate of every cell, barcode diversity is largely preserved (165 of 492; Fig. 4e). This is the signature that distinguishes the two origins in barcoding experiments<sup>2,6</sup>.

Disrupting expression memory changes the outcome, and when it is disrupted matters. Accelerating promoter switching 20-fold from two cell-cycle times before the drug, as memory-disrupting epigenetic compounds do<sup>9</sup>, does not reduce the number of surviving clones if the acceleration stops when the drug arrives: it raises them, from 98 to 120 (mean of 5 seeds each,  $P = 0.009$ , Welch's t test). Reshuffling spreads the resistant state over

more lineages and moves the resistant fraction towards its stationary value (0.09 to 0.11), so more clones carry a protected cell when the drug arrives, even though protection itself is untouched. Maintaining it during exposure reduces the count to 7 clones (10 cells, against 792; Fig. 4f), because resistant cells now leave the protected state within a fraction of the time they need to found a colony. In the model, memory must be disrupted while the drug is present. Compartment theory for phenotypic switching already predicts that an agent which perturbs the retention of the resistant state can turn treatment failure into success<sup>28</sup>; what the simulations add is the timing requirement and a lineage-resolved readout, the number of surviving clones, by which it can be tested. A transient pulse enriches high-expressing cells and, with slow switching, the enrichment persists for many generations after the drug is removed, whereas with fast switching it relaxes within a few cycles (Fig. 4g): phenotypic selection with stable inheritance, as in the original MGMT model<sup>29</sup>. Finally, scanning release periods and doses of intermittent schedules (Fig. 4h), the most effective schedule is release period 0, dose 2 (growth rate +0.004 per time unit) for pre-existing tolerance, release period 10, dose 2 (growth rate -0.016 per time unit) when the resistant state carries a fitness cost, and release period 10, dose 1 (growth rate +0.009 per time unit) for drug-induced tolerance. Without a fitness cost nothing beats continuous dosing at the highest dose, since every release period lets the resistant subpopulation regrow (the drug does not act on the switching rates in this model; when it also stabilises the resistant state, as in the melanoma case study below, holidays regain a small advantage even without a cost). With a fitness cost the population declines under every schedule at dose  $\geq 0.5$ , and release periods of up to 10 time units match continuous dosing (-0.013 per time unit under continuous dose 2), because cells that leave the resistant state during a holiday are killed on re-exposure. For drug-induced tolerance the ranking of doses reverses: the highest dose induces tolerance in more cells than it kills, so dose 1 outperforms dose 2 at every release period (release period 10, dose 1 (growth rate +0.009 per time unit), against +0.012 for continuous dose 2), the intermediate-dose optimum reported previously<sup>30</sup>; the benefit of release periods reported in that work does not appear here, because in this model the induced state decays at the same rate with and without drug, so a holiday only postpones the death of reverting cells while all cells grow.

The hazard above does not depend on where a cell is in its division cycle, and the calibration below will show that fate data cannot exclude one that does, so we repeated these simulations with three quarters of the hazard confined to a window around mid-cycle. Population decay still rises with dose (-0.001 to +0.023 per time unit over doses 0.5 to 2, against +0.023 to +0.050 without the gate) and the mean time to death stays nearly dose-invariant (8.4 to 9.9 against 6.2 to 8.5 time units), so neither signature depends on a cycle-blind hazard. Sisters share a birth time and so a cycle phase, which lets a gated hazard correlate their fates with no inherited state; it does not do so to any useful degree, lifting the sister concordance of the memoryless fast-switching control only to +0.008 above independence (from +0.000), against +0.054 for the memory gene, so the kin signature still reads expression memory rather than the cycle. The schedule conclusions are the fragile ones, because the best dose is unchanged in 3 of 3 models, but the best release period is not. With a fitness cost of resistance and with drug-induced tolerance, the holidays that won without the gate collapse to continuous dosing, because a gated hazard kills the reverting cells more slowly on re-exposure (Supplementary Note 13 and Supplementary Fig. 5).

## Simulated ground truth benchmarks inference methods

Because every cell's clone, lineage and true state are known, the framework supplies ground truth that observational data cannot (Fig. 5). For 40 independent genes whose memory times span three decades, MemorySeq-style clonal variance scores computed on true concentrations rise from 1.0 for genes with memory far below the cell-cycle time to 29.6 for the slowest genes, and the ranking survives computation on sequenced counts at 20% capture efficiency (Fig. 5a); the score separates genes with memory longer than the cycle from the rest with an area under the receiver operating characteristic (ROC) curve of 1.00 on true concentrations and 0.99 on sequenced counts.

For a 30-gene network simulated at fixed volume, in a dividing population and after sequencing, correlation-based and GENIE3 inference lose most of their accuracy in the dividing population when raw counts are used (area under the precision-recall curve 0.52 to 0.14 for Pearson correlation and 0.32 to 0.13 for GENIE3, against 0.10 for a random ranking of undirected pairs and 0.05 for directed edges), because volume and gene copy number induce positive correlations between all genes. Normalising counts to concentrations removes the volume effect but recovers little (0.18 and 0.18), since the dependence on copy number and cell age remains; regressing expression on the true cell-cycle covariates, which the simulation provides and which cell-cycle scoring approximates in real data, recovers most of the loss (0.46 and 0.30). Sequencing at 15% capture efficiency lowers accuracy further (0.11 raw and 0.24 library-size-normalised for Pearson correlation; 0.07 and 0.08 for GENIE3; Fig. 5b). Imputing the sequenced counts before inference recovers part of that loss: MAGIC raises the area under the precision-recall curve from 0.11 to 0.32 for Pearson correlation and from 0.07 to 0.18 for GENIE3, and kNN-smoothing gives 0.16 and 0.12 (Fig. 5c). Neither reaches the accuracy obtained at fixed volume, nor that of regressing out the cell-cycle covariates, so imputation mitigates the sequencing loss without touching the growth confound that produces most of it<sup>25</sup>. Knocking down the five most connected regulators of a 20-gene network gives true fold changes that a no-change baseline predicts with a mean coefficient of determination of -0.16. Propagating observational correlations beats that baseline for 4 of 5 knockdowns, with coefficients of determination from 0.09 to 0.43, and fails on the remaining one (-2.39), so its mean over the five is -0.23 against a median of 0.30 (Fig. 5d). A predictor that is useful for most perturbations and badly wrong for one is precisely what an out-of-distribution evaluation has to expose, and exposing it needs the causal ground truth that the simulation supplies<sup>13,14</sup>.

## Inference recovers parameters and exposes division bias

On a snapshot of 2,000 non-dividing cells, ABC-SMC with log-uniform priors over three decades and the Wasserstein distance between simulated and observed counts<sup>31</sup> concentrates on the true telegraph parameters and agrees with the exact maximum-likelihood estimate (posterior medians (0.31, 0.60, 20.9) for truth 0.3, 0.6, 20; Fig. 6a), the broadest posterior falling on the promoter inactivation rate, which a single snapshot constrains least. On a snapshot of a growing and dividing population with volume-scaled transcription and mid-cycle replication, the naive telegraph fit is biased ( $k_{\text{on}} = 0.47$ ,  $k_{\text{off}} = 1.95$ ,  $k_{\text{tx}} = 71.5$ ) because volume and copy number inflate counts and their dispersion, whereas ABC with the population simulator as forward model recovers the truth ((0.30, 0.68, 20.8); Fig. 6b). Drug parameters follow from population data in the same way: from a kill curve and the concordance of sister fates, the posterior over the maximal death rate and the protection threshold is a ridge (weighted correlation of the two log parameters -0.85) that contains the truth (Fig. 6c), and posterior predictive kill curves simulated at the posterior median reproduce the observed one (Fig. 6d). The two summaries constrain the product of death rate and unprotected fraction, so separating the parameters requires measuring the protective protein: an experimental-design question the forward model answers before an experiment is run (the tolerance and acceptance-rate schedules of all three ABC runs are in Supplementary Fig. 2).

## Calibration to time-lapse data is tested on held-out data

A simulator is useful for treatment questions only if it can be fitted to the data a laboratory produces and then predict what it was not fitted to. We therefore calibrated the resistance-gene model to time-lapse measurements in which U2OS cells were tracked for three days under 7, 10 and 13  $\mu\text{M}$  cisplatin and every cell present at drug addition was scored as dying, dividing or surviving without dividing<sup>6</sup> (Supplementary Table 4). Six parameters, the two promoter switching rates, the maximal death rate, the  $\text{EC}_{50}$  and Hill coefficient of death and the  $\text{IC}_{50}$  of growth arrest, were fitted to the fate fractions at 7 and 13  $\mu\text{M}$  by simulated moments with common random numbers, under a weak prior keeping the memory of the resistant state near the two to three generations that

the same study inferred from lineage correlations (Methods). The calibrated model reproduces the training fractions with a root-mean-square error of 0.071 (memory 2.8 generations, resistant fraction 15.2%,  $EC_{50}$  16.7  $\mu$ M; Fig. 7a,f). That error should be read against the precision of the measurement: the observed fractions carry a mean binomial standard error of 0.028 and repeated simulations differ by 0.015, so the fit is about 2.2 times the combined noise and the model does not reproduce the training data within error. At the held-out concentration the root-mean-square error is 0.093, and the discrepancy is specific rather than diffuse: the model matches the fraction surviving without dividing but over-predicts death and under-predicts division (observed died, divided and survived 0.28, 0.31 and 0.41; predicted 0.39, 0.20 and 0.41; Fig. 7a), so it captures the ordering of the three concentrations while misplacing the split between death and division at the intermediate one.

Six parameters fitted to six numbers are not all determined by them. Moving one parameter at a time from the fitted values, with the others held there, the fate fractions constrain the death parameters to within about a factor of 2 but leave the two switching rates free over roughly a factor of 6 (Supplementary Fig. 3a), which is why the memory of the resistant state was given a prior rather than left to the fate data. Holding the death parameters at their fitted values, memories from 1.0 to 9.0 generations paired with resistant fractions from 1% to 25% all describe the fate fractions within the noise of the measurement (Supplementary Fig. 3b), because a shorter memory with more resistant cells and a longer memory with fewer are hard to tell apart from fates alone. The parameters quoted above should therefore be read as one member of a family of fits that describe these fate fractions equally well, and the biological reading of the memory rests on the lineage correlations that set the prior, not on the fate fractions. Halving the integration step changes the simulated fractions by 0.005, less than the seed-to-seed spread, so the discretisation is not a factor (Supplementary Fig. 3c).

A second mechanism can produce the same fates. Cisplatin kills preferentially during replication, so where a cell sits in its division cycle when the drug arrives is a competing explanation both for the dose-invariance of death timing and for the cells that survive three days without dividing: a hazard confined to part of the cycle spares whatever never enters it, with no heritable state at all. We therefore refitted the model with a cell-cycle-dependent hazard, adding one parameter, the fraction of the hazard that is independent of cycle position, with killing otherwise peaking at mid-cycle, under the same objective, bounds, budget and seeds (Methods). The two fits describe the training fractions equally well (0.064 against 0.061, against a mean sampling error of 0.028 in the measurement), and the cycle-dependent fit does not improve the held-out prediction either (0.106 against 0.095). The fitted cycle-independent fraction is 0.34, but values from 0.20 to 0.80 all sit within one sampling error of the best fit, so the fate fractions do not determine it (Supplementary Note 12 and Supplementary Fig. 4). Three fate counts per concentration cannot separate inherited protection from cell-cycle position; the measurement that can is the cycle position of each cell at the moment the drug arrives, which a cycle reporter records directly.

Properties that were not fitted can then be compared with the data. The predicted kill curves (Fig. 7b) are 1.5-fold steeper at 13  $\mu$ M in their first two days than in their second half, but they do not plateau, as HCT116 cells do after about 100 h. Death fates are correlated between sisters (0.09), first cousins (0.06) and second cousins (0.05), and the correlation decays to 0.02 for third cousins and 0.00 for unrelated cells (Fig. 7c), the pattern reported for HCT116 lineages. The mean time to death changes by 8% between 7 and 13  $\mu$ M while the death fraction rises from 0.13 to 0.59 (Fig. 7d), the dose-invariance of single-cell timing that motivated the original study, and it arises here because dose changes the fraction of unprotected cells rather than their kinetics. The plateau the kill curves miss is the clearest failure of the minimal model: reproducing it requires cells to stay in the resistant state for longer than the memory of 2.8 generations allows, and that memory was set by the prior taken from the lineage correlations rather than by the kill curves. The two observations therefore pull in opposite directions within this minimal model: correlations that vanish by the third cousin call for a short memory, and a

kill curve that flattens calls for a long one. A single switching gene cannot satisfy both, which is a concrete and testable limitation, and points to a drug-induced stabilisation of the resistant state, for which there is independent evidence<sup>2</sup>, as the mechanism a lineage-tracing experiment should look for. The calibration landscape (Fig. 7e) shows how the fitted death rate and EC<sub>50</sub> are constrained by the fate fractions alone; the protocol applies to any observable a laboratory records.

## Schedule optimisation is consistent with clinical outcomes

With a calibrated model the framework becomes a schedule testbed. We first asked which schedule best delays regrowth of a melanoma-like population with a rare, heritable pre-resistant state<sup>2</sup> treated at a fractional-killing dose, comparing continuous dosing, the intermittent schedule of the SWOG S1320 trial, in which patients received a continuous 8-week lead-in and were then randomised to continuous dosing or to cycles of 3 weeks off and 5 weeks on<sup>32</sup>, and adaptive therapy, in which the drug is withdrawn when the population has halved and resumed when it regrows to its initial size<sup>33</sup>. Because adaptive therapy deliberately lets the tumour regrow, schedules are compared by the time after randomisation at which the population exceeds 120% of its pre-treatment size (loss of control), with the RECIST-like time to progression from the nadir reported alongside (Methods).

The answer depends on two properties of the resistant cells (Fig. 8a–d). When the resistant state is fully protective and carries no fitness cost, as lineage data indicate for pre-existing tolerance<sup>6</sup>, resistant cells grow at the same rate with or without drug, holidays only let sensitive cells regrow and be killed again, and the three schedules lose control after 23, 26 and 27 weeks, the intermittent schedule delivering 65% of the cumulative dose. When resistant cells pay a 50% fitness cost in the absence of drug, a coarse stand-in for the drug dependence that made intermittent dosing superior in vemurafenib-treated patient-derived xenografts, where withdrawal drove established resistant tumours to regress rather than merely grow more slowly<sup>34</sup>, holidays let sensitive cells outcompete them: none of the three schedules loses control within the 52-week follow-up (52+, 52+ and 52+ weeks, censored), and adaptive therapy achieves this with 36% of the cumulative dose of continuous dosing. When resistant cells are only partly protected and grow at half speed under drug, holidays release them: continuous dosing keeps control for 52+ weeks against 49 for the intermittent schedule and 43 for adaptive therapy, the ranking observed in the trial, where continuous dosing gave a median progression-free survival of 9.0 months against 5.5 months for intermittent dosing. Gating three quarters of the death hazard to a window of the cell cycle, as in the persister simulations above, leaves these comparisons standing (Supplementary Note 13 and Supplementary Fig. 5): the best schedule is unchanged in 3 of 3 mechanisms and the model still favours continuous dosing under partial protection (52 against 43 weeks, from 52 and 49). That the benefit of holidays should depend on a cost of resistance is established by compartment models of adaptive therapy, which also show that tumour turnover modulates how large that cost must be<sup>35</sup>. Here the dependence emerges from single cells whose resistance is a heritable expression state rather than a fixed compartment, the cost and the degree of protection are properties of cells that lineage experiments measure, and the third case shows that partial protection reverses the ranking even when a cost is absent. The optimiser makes the dependence explicit (Fig. 8e): over intermittent schedules of any period and duty cycle after the lead-in, the best schedule is a period of 2.2 weeks with 39% of the time on drug (27 weeks to loss of control) without a fitness cost, a period of 1.0 weeks with 47% of the time on drug (52 weeks to loss of control) with a fitness cost and a period of 1.0 weeks with 100% of the time on drug (52 weeks to loss of control) with partial protection. The model therefore reproduces both the clinical and the preclinical outcome and shows their difference to depend on measurable properties of the resistant cells, their growth under drug and their fitness without it, both properties of cells rather than of the tumour, which growth and lineage experiments can bound, within the identifiability limits of Fig. 6c, before a schedule is chosen. The mechanism is set in each case, not inferred from the outcome, so this is a demonstration that the ranking follows from cell properties rather than a

claim that these particular cells carry them; and because the trial and the xenograft also differ in drug and species (dabrafenib plus trametinib in patients, vemurafenib monotherapy in mice), their contrast is not attributable to resistant-cell fitness alone.

The second case study concerns glioblastoma, where the RTOG 0525 trial compared the standard temozolomide regimen (150–200 mg/m<sup>2</sup> on days 1–5 of 28-day cycles) with a dose-dense regimen (75–100 mg/m<sup>2</sup> on days 1–21) that delivers about twice the cumulative dose and was designed to deplete MGMT, and found no difference in survival for either MGMT-methylated or unmethylated tumours<sup>36</sup>. We simulated both regimens with one-compartment pharmacokinetics (elimination half-life 2.1 h; ref.<sup>37</sup>) on populations in which 1% (methylated) or 30% (unmethylated) of cells express MGMT, with MGMT protein protecting cells from the drug<sup>29</sup> and either stable or consumed by it stoichiometrically, one molecule per lesion repaired, as a suicide enzyme is, so that the pool follows the cumulative exposure rather than the peak concentration (Fig. 8f,g). With stable MGMT, the depth of response over six cycles is 0.20 log<sub>10</sub> units for the standard regimen and 1.00 for the dose-dense regimen in the methylated population (0.08 and 0.10 in the unmethylated one), and a 21-day regimen at the standard cumulative dose achieves no measurable kill (0.00 log<sub>10</sub> units): the death hazard rises steeply with concentration, so spreading the same dose over more days loses efficacy, and the advantage of the dose-dense arm in the model is entirely its larger cumulative dose. With stable MGMT the two arms therefore differ little in the unmethylated population (0.08 against 0.10 log<sub>10</sub> units), as in the trial. Consumption changes that. Because one molecule is spent per lesion repaired, the pool follows the cumulative exposure, which the dose-dense regimen more than doubles: in the methylated population its response deepens to 1.52 log<sub>10</sub> units against 0.22 for the standard regimen, and in the unmethylated population, where 30% of cells start protected, the same depletion turns a regimen that barely worked into one that controls the population (1.08 against 0.18 log<sub>10</sub> units; final size 0.1 against 1.1 times the initial size, and 4.9 and 3.6 with stable MGMT). Modelled as the suicide enzyme it is, MGMT therefore predicts a clear dose-dense advantage in both populations, which RTOG 0525 did not find. Read as a bound rather than as a fit, that disagreement is informative: either temozolomide consumes MGMT in tumours far more slowly than this stoichiometry allows, or the benefit of depleting it is offset by something the model leaves out. At the standard cumulative dose the number of dosing days per cycle that minimises the final population is 9 (methylated) and 9 (unmethylated) with stable MGMT, and 9 and 5 when MGMT is consumed (Fig. 8h). These simulations are not fits to the trial, whose end points depend on much that the model omits, dose-limiting toxicity and the dose reductions it forces among it. What they add is a sharper version of the rationale for dose-dense dosing: the benefit it is supposed to deliver exists in the model only at consumption rates that the trial’s null result appears to exclude, so the rate at which temozolomide consumes MGMT in the tumour is both the quantity the argument rests on and the one that has not been measured.

## Discussion

Biomodelling.jl 2.0 fills a gap in the single-cell toolkit: a simulator in which expression memory, cell-size effects, cell-cycle dependence and drug-tolerant persistence follow from explicit biophysics instead of being assumptions of a generative model. It is worth separating what is new here from what it confirms. The individual ingredients are not new: stochastic kinetics in growing and dividing cells have been simulated before<sup>23–25</sup>, lineage-tracing data have their own simulators<sup>21,26</sup>, compartment models of adaptive therapy established that the benefit of treatment holidays depends on a cost of resistance and on turnover<sup>33,35</sup>, and phenotypic-switching theory established that perturbing the retention of a resistant state can decide a treatment outcome<sup>28</sup>. Three things are new. First, the combination itself: molecular kinetics, volume-scaled transcription, gene replication, partitioning and promoter inheritance, drug schedules with state-dependent killing, a complete lineage record, observation models and

inference in one forward model, which is what allows a resistance phenotype to be a heritable expression state with measurable kinetics rather than a compartment with an assumed rate. Second, the ground truth this makes available: the same simulation supplies known memory genes, known networks, known perturbation effects and the true lineage tree, which is what the quantification of how growth and sequencing destroy network inference, and of how far cell-cycle regression recovers it, depends on. Third, the calibrate-then-predict protocol applied to schedule questions, which turns a mechanism into a falsifiable prediction: here it identifies the growth of resistant cells without drug and their residual growth under drug as the two properties that decide whether holidays help, and points to the rate at which temozolomide consumes MGMT as the measurement that would separate the two glioblastoma regimens. That difference has practical consequences. As a hypothesis engine for non-genetic drug tolerance, the framework answers which schedule, which combination or which memory-modulating pretreatment minimises the long-term fitness of a treated population, with cell-level rules that can be matched to time-lapse and lineage-tracing data. As a ground-truth generator, it produces data with known memory genes, known networks, known perturbation effects and known lineage trees, which is what MemorySeq-style scores, lineage-aware models and virtual-cell benchmarks need in order to be evaluated fairly, and what existing simulators cannot supply. As a likelihood-free inference engine, it estimates mechanistic parameters from data that are now routinely collected, while quantifying the bias of simpler models.

The calibration and schedule studies set out a workflow for in silico single-cell pharmacology: declare the training and validation observables before fitting, fit by simulated moments or approximate Bayesian computation with common random numbers, validate on conditions and statistics that were not fitted, and only then compare regimens, reporting outcomes that map onto clinical end points. The case studies show why mechanism matters. The same question, continuous or intermittent dosing, has opposite answers depending on two properties of the resistant cells, their growth under drug and their fitness without it, and a trial result and a xenograft result that appeared to contradict each other are both reproduced once those properties are set — set in each case rather than inferred from the outcomes, so the case studies show that the ranking follows from cell properties, not that any particular tumour has them. Because the framework simulates the lineage-resolved data from which such properties are measured, it can also say which experiment would settle the question before a trial is designed.

Several limitations delimit this release. Cells are well-mixed compartments without spatial structure or cell-cell signalling; the pharmacology is a one-compartment model; growth is exponential with phenomenological size control; and the inference layer, although general, relies on user-chosen summary statistics, so posterior calibration should be checked with predictive simulations. The persister case study is deliberately minimal, a single resistance gene with slow switching, and its calibration used fate fractions transcribed from a published study rather than raw tracks. Its death hazard was first written without cell-cycle dependence, which matters for a drug such as cisplatin whose lethality is enhanced during replication. Refitting with a cycle-dependent hazard shows that the two mechanisms are not separable by these data: a model in which killing is confined to the replication window accounts for the fate fractions as well as heritable protective expression does, and the fraction of the hazard that is cycle-independent is not determined by them at all. Cell-cycle position at the moment the drug arrives is therefore a live alternative to inherited protection, and the experiment that distinguishes them is a cycle reporter read at drug addition rather than more fate counts. Six parameters fitted to six fractions leave the switching rates weakly determined, so the memory of the resistant state rests on the lineage correlations used as a prior rather than on the fates themselves; and that memory is carried by one slow promoter, a coarse proxy for chromatin-based inheritance whose single-exponential, symmetric form is a simplification of a process that can be multi-timescale and bistable. The clinical case studies use literature values for cell-cycle times, resistant fractions and pharmacokinetics rather than patient-specific measurements, so their conclusions concern mechanisms and rankings, not absolute survival times. In the glioblastoma study MGMT is consumed stoichiometrically, one

390 molecule per lesion repaired, so its depletion follows the cumulative exposure as a suicide enzyme’s should; what  
 391 remains unmeasured is the rate constant of that consumption, which is why we name it as the quantity that  
 392 separates the regimens. MGMT is in any case only one determinant of a temozolomide response that is also  
 393 mismatch-repair-dependent, and the methylated and unmethylated MGMT fractions are illustrative values that  
 394 bracket low and high expression rather than measured quantities. The memory of the resistant state is expressed  
 395 in these tumour-scale studies as a number of net population doublings, not of cell divisions, and so is not the  
 396 division-counted memory of memory-sequencing experiments. A bridge to the Catalyst.jl ecosystem is not part of  
 397 this release. The benchmark results should be read as demonstrations of what the framework makes possible  
 398 rather than as verdicts on the methods compared, which were run with default settings on small networks.

399 Spatial coupling through neighbourhood-dependent rates, cell-cell communication, chromatin states with their  
 400 own inheritance rules and neural simulation-based inference are natural extensions of the existing forward model.  
 401 Because the framework is deterministic given a seed, fast enough to simulate tens of thousands of cells in minutes,  
 402 and writes AnnData files, it can be dropped into single-cell analysis pipelines as a source of mechanistically  
 403 grounded synthetic data, and into virtual-cell training and evaluation as a source of causal, out-of-distribution  
 404 contexts.

## 405 Methods

### 406 Reaction model and propensities

407 A model is a set of species and reaction channels. Species are molecules (partitioned at division) or promoter  
 408 states (inherited). For a reaction with reactant counts  $x_i$ , stoichiometric coefficients  $\nu_i$ , rate constant  $k$  and  
 409 volume exponent  $e$ , the mass-action propensity is

$$a = k V^e \prod_i \binom{x_i}{\nu_i},$$

410 where by default  $e = 1 - \sum_i \nu_i$ , so that zero-order production scales with volume, first-order reactions are volume  
 411 independent and bimolecular reactions scale as  $1/V$ ; this keeps concentrations invariant under growth. Regulated  
 412 reactions multiply this by  $b + (1 - b)F(\mathbf{c})$ , where  $b$  is a basal fraction and  $F$  combines Hill terms of the regulator  
 413 concentrations  $c_j = x_j/V$ ,  $h_j = c_j^{n_j}/(K_j^{n_j} + c_j^{n_j})$  for activators and  $1 - h_j$  for inhibitors, either as a product  
 414 (AND logic) or as  $1 - \prod_j (1 - h_j)$  (OR logic). Promoter switching is represented by explicit promoter species (for  
 415 instance  $G_{\text{off}} \rightleftharpoons G_{\text{on}}$ ) grouped into promoter groups. Custom propensities can be supplied as functions of counts,  
 416 volume, parameters and time.

### 417 Stochastic simulation kernels

418 Within one update step the cell volume and gene copy number are held fixed and the reaction network is advanced  
 419 by one of four kernels: the direct method<sup>38</sup> with propensities updated through a species-reaction dependency  
 420 graph; fixed-step Poisson tau-leaping<sup>39</sup>; a hybrid scheme in which a leap that would drive any species negative is  
 421 discarded and that interval is simulated exactly instead (the “switching” scheme of version 1, with the discarded  
 422 interval resolved exactly rather than clipped); and adaptive tau-leaping with the step-size selection of<sup>40</sup> and  
 423 the critical-reaction partition of<sup>41</sup>, falling back to a fixed number of exact events when the proposed leap is  
 424 shorter than  $10/a_0$ . The hybrid scheme is approximate rather than exact: a leap is kept only when it leaves  
 425 every species non-negative, so an accepted step follows the Poisson leap truncated to the non-negative region,  
 426 and only the discarded intervals follow the exact process. The discrepancy falls with  $\tau$  and is bounded by the

427 Kolmogorov-Smirnov test of Fig. 2d, which the scheme passes at  $\tau = 0.05$  and fails at  $\tau = 0.2$  for the models  
 428 tested; the direct method is used wherever exactness is required (Supplementary Table 1 lists the kernel correctness  
 429 tests). Every kernel takes an explicit random number generator.

## 430 Cells, growth, replication and division

431 Each cell carries its volume  $V$ , age, generation, lineage identifiers, gene copy state, growth rate, molecule counts,  
 432 a cell-specific parameter vector and its own random number generator (seeded from the parent's generator, which  
 433 makes results independent of the number of threads). Volume grows exponentially,  $\dot{V} = \lambda V$ , with an optional  
 434 log-normal spread of  $\lambda$  between cells. Division is triggered by a sizer ( $V \geq V_{\text{div}}$ ), an adder ( $V - V_{\text{birth}} \geq \Delta$ ) or  
 435 a timer (age  $\geq T$ ), each with a per-cell noise factor drawn at birth. At division the first daughter inherits a  
 436 volume fraction  $f \sim \mathcal{N}(0.5, \sigma^2)$  (clipped to  $[0.05, 0.95]$ ) and each molecule is assigned to it independently with  
 437 probability  $f$  (binomial partitioning); a beta-binomial option models over-dispersed partitioning. Promoter groups  
 438 are inherited by both daughters when unreplicated and split one copy per daughter, in random assignment, when  
 439 replicated. Gene replication, if enabled, doubles the promoter counts (and the propensity of reactions flagged as  
 440 copy-number dependent) once the cell has completed a set fraction of its cycle, and is reset at division. Populations  
 441 can be started from per-founder states and volumes (for instance a snapshot of a burnt-in constant-size population).  
 442 Population control is either constant size (every second daughter replaces a uniformly chosen cell, the behaviour  
 443 of version 1), free growth (all daughters kept, with uniform subsampling above a cap and book-keeping of the true  
 444 size) or logistic (free growth with an additional hazard  $\lambda N/K$ ).

## 445 Drug and perturbation layer

446 A perturbation combines a dose schedule  $d(t)$  (constant, pulsed, piecewise, one-compartment bolus pharmacoki-  
 447 netics or an arbitrary function) with effects evaluated for every cell at every step: a death hazard

$$h = h_{\max} \frac{d^m}{\text{EC}_{50}^m + d^m} \cdot \frac{K^q}{K^q + c_P^q},$$

448 where  $c_P$  is the concentration of a protective species (survival probability over a step is  $e^{-h \Delta t}$ ); growth inhibition  
 449 by  $1/(1 + (d/\text{IC}_{50})^m)$ ; a dose-independent fitness cost of a cell state,  $1 - c_{\max} c_P^q/(K^q + c_P^q)$ , on the growth rate;  
 450 multiplicative modulation of any parameter by a function of the dose; and genetic perturbations (knockdown,  
 451 knockout, over-expression) of any parameter in a random fraction of cells, the flag being inherited by daughters.

452 Two further effects couple the drug to the cell cycle and to the protein that resists it. A cell-cycle-dependent  
 453 hazard multiplies  $h$  by

$$\beta + (1 - \beta) \exp \left[ -\frac{1}{2} \left( \frac{\varphi - \varphi_0}{w} \right)^2 \right],$$

454 where  $\varphi$  is the cell's progress through its division cycle (0 at birth, 1 at division), so that killing peaks at  $\varphi_0$  and  
 455 falls to a fraction  $\beta$  of the peak elsewhere;  $\beta = 1$  is the cycle-independent hazard and  $\beta = 0$  confines killing to a  
 456 window of width  $w$  around  $\varphi_0$ . This represents an agent whose lesions are converted into death during replication.  
 457 Stoichiometric consumption of a protective protein removes molecules of it at rate

$$k d V \frac{c_P}{c_P + K_m},$$

458 by a Poisson draw over the step. Because lesions form in proportion to the dose, the rate is set by lesion  
 459 formation alone while the protein is abundant, so the pool follows the cumulative exposure rather than the peak

concentration, and falls with what remains once the pool runs low; resynthesis is whatever the reaction model provides. This is the behaviour of a suicide enzyme such as MGMT, which is inactivated by the very lesion it repairs.

## Lineage statistics

The lineage table records, for every cell, its parent, founder (clone), generation, birth and end times, fate (divided, died, removed, alive) and its volume and state at birth and at the end of its life. From it we compute Pearson correlations of expression at division between mothers and daughters, sisters and cousins; the lineage autocorrelation  $r_g$  between a cell and its ancestor  $g$  generations earlier, and the memory timescale  $\tau$  from  $r_g = e^{-g/\tau}$ ; single-lineage traces by following the ancestral line of a surviving cell; the coefficient of variation squared of concentration across the population versus along single lineages; Luria-Delbrück fluctuation tests (variance of a phenotype across clones grown from single cells compared with the binomial expectation); and MemorySeq-style clonal scores<sup>42</sup>, the variance of clone-mean expression divided by its permutation-null expectation.

## Observation models

True counts are converted to observations by a scRNA-seq model in which each cell captures every molecule independently with a Beta-distributed efficiency (mean and coefficient of variation set by the protocol), optional multinomial down-sampling to a Poisson-distributed depth, optional additional dropout and batch scaling; an smFISH model with a fixed detection efficiency; and a time-lapse model that samples a species along lineages at a fixed interval with additive or multiplicative measurement noise. Because counts scale with volume, library size correlates with cell size, reproducing the transcriptome-size confound of real data. Outputs are written as CSV, as AnnData-compatible HDF5 files, or as Newick lineage trees.

## Inference

Snapshot counts of a gene in non-dividing cells follow the Beta-Poisson law of the telegraph model<sup>27</sup>, which we evaluate through Kummer’s confluent hypergeometric function with rescaled series summation, and maximise by a grid-initialised Nelder-Mead search in log-parameter space. For any model, including populations, we implement sequential Monte Carlo approximate Bayesian computation<sup>43</sup> with an adaptive tolerance schedule (the  $\alpha$  quantile of the previous generation’s distances), Gaussian perturbation kernels of variance twice the weighted population variance, importance weights against the prior, and parallel evaluation of the user-supplied distance function; for count distributions we use the Wasserstein-1 distance between simulated and observed samples<sup>31</sup>.

## Calibration to laboratory data and validation strategy

Reference values were transcribed from the text and tables of the cited articles into `paper/data/` with their provenance (Supplementary Table 4). For the cisplatin case study the model is a resistance gene with slow promoter switching whose protein product protects against a dose-dependent death hazard and against a dose-dependent growth arrest (Methods above), with a 24-h cell cycle (at the fast end of reported U2OS doubling times and consistent with the intermitotic times tracked by<sup>6</sup>). Its six free parameters (activation and inactivation rates, maximal death rate,  $EC_{50}$  and Hill coefficient of death,  $IC_{50}$  of growth arrest) were fitted by the method of simulated moments: the root-mean-square difference between simulated and observed fractions of cells that died, divided or survived without dividing during 72 h of drug, for the training concentrations (7 and 13  $\mu$ M), plus a weak penalty keeping the memory of the resistant state near the two to three generations inferred from lineage correlations, was minimised with the bounded optimiser described below using common random numbers (the

same seeds for every candidate). Validation used only quantities not fitted: the fate fractions at the held-out concentration (10  $\mu\text{M}$ ), the biphasic kill curve and its plateau, the extent of lineage correlations in fate (sisters, first, second and third cousins) and the invariance of single-cell division and death times across concentrations. The protocol is deliberately generic: any observable computable from a `PopulationResult` (kill curves, fate fractions, lineage correlations, snapshot distributions) can enter the distance, and the split between training and validation conditions is declared before fitting.

## Clinical regimens, treatment outcomes and schedule optimisation

Regimens are built from the schedule primitives of the perturbation layer: `PulsedDose` with a finite number of cycles, `PiecewiseDose` for arbitrary calendars (the S1320 schedule), `daily_boluses` and `cycle_days` for daily oral doses with one-compartment pharmacokinetics, and `AdaptiveDose`, a feedback schedule that withdraws the drug when the population falls below a fraction of its initial size and resumes it when the population regrows above another fraction, evaluated with the current population size at every step<sup>33</sup>. Outcomes computed from a simulation are the net growth rate, the depth of response ( $\log_{10}$  of the ratio of the initial to the smallest population), the time to progression (first time the population exceeds a threshold, 1.2, times its running minimum, the analogue of RECIST progression from the nadir; censored at the end of the simulation), the extinction probability over replicates and the cumulative dose actually applied. `optimize_schedule` minimises any such outcome over a box of schedule parameters: a log-spaced grid (or a Latin hypercube when the grid is large) is evaluated with the same random seeds for every candidate, and the best point is refined by Nelder-Mead in a bounded transform of the box; constraints such as a fixed cumulative dose enter as penalties or as the parameterisation (for instance a per-day dose equal to the budget divided by the number of dosing days). The case studies are run at the scale of a tumour rather than a culture: the cell cycle of the model is the net doubling time of the tumour (4 weeks for melanoma, 40 days for glioblastoma). The glioblastoma value lies within the volume doubling times measured for untreated tumours, a median of 21 days in one magnetic-resonance series and 50 days in another<sup>44,45</sup>; the melanoma value is an assumption, since net doubling times of treated cutaneous melanoma are not established, and it sets the time axis of the melanoma case study rather than the ranking of schedules, which follows from the resistance mechanism. Growth is exponential throughout, which is a simplification: the same measurements favour Gompertzian slowing in large glioblastomas<sup>45</sup>, so absolute regrowth times in Fig. 8 are shorter than a saturating model would give. Gene-expression rates are scaled to the hour (mRNA half-life 7 h and protein half-life 35 h, generic mammalian values; at the tumour time scale both species turn over many times per generation, so the results are insensitive to them and all heritable memory resides in the promoter), and founders are drawn from the stationary distribution of promoter states. The melanoma case uses a pre-resistant fraction of 1:200 (a per-marker frequency<sup>2</sup> used as a proxy for the functional pre-resistant fraction, which is not the same quantity), with a memory of five net doublings (the model's generation is the net doubling of the tumour, a modelling choice rather than the cell-division-counted memory of memory sequencing<sup>42</sup>) and drug effects at the fractional-killing dose: sensitive cells arrest and die with a half-life of two weeks, cells in the resistant state are protected, and the drug stabilises the resistant state ten-fold (the drug-induced reprogramming of<sup>2</sup>). Three variants of the resistant state are compared: full protection without cost; a 50% fitness cost in the absence of drug<sup>34</sup>; and partial protection, in which resistant cells grow at half speed under drug. Progression is scored in two ways: from the nadir, as in RECIST (a 20% increase in diameter, taken as a 1.73-fold increase in cell number, after the 8-week lead-in; a well-mixed cell count cannot represent RECIST's absolute 5 mm threshold, its new-lesion and non-target criteria or its sum-of-diameters basis, and it assumes cell number proportional to volume, so this is an analogue rather than the clinical end point), and as loss of control, the time after randomisation at which the population exceeds 120% of its pre-treatment size, the end point of adaptive-therapy studies, which is the one used to compare

schedules that deliberately let the tumour regrow. The temozolomide case uses a one-compartment elimination half-life of 2.1 h<sup>37</sup> (a plasma value from a three-compartment study; the brain-tumour compartment sees roughly a fifth of the plasma exposure with slower kinetics, so this is a surrogate), the regimens of RTOG 0525<sup>36</sup> (200 mg/m<sup>2</sup> on days 1-5 or 100 mg/m<sup>2</sup> on days 1-21 of 28-day cycles, six cycles), MGMT-expressing fractions of 1% and 30% for methylated and unmethylated promoters (illustrative values bracketing low and high expression rather than measured fractions; methylation shifts graded expression rather than a bimodal positive fraction), and MGMT either stable or consumed stoichiometrically by the drug, one molecule per lesion repaired ( $k = 300$ ,  $K_m = 150$  in the expression above). The rate constant was chosen so that the consumption rate at the standard bolus peak equals that of the dose-scaled first-order degradation used in an earlier version of this model, which isolates the consequence of letting depletion follow the cumulative exposure instead of the peak concentration; this is the suicide-enzyme behaviour that motivated dose-dense schedules.

## Random regulatory networks

Random networks are built with a chosen number of activating and inhibiting edges under Erdős-Rényi or preferential-attachment (scale-free) sampling; regulation acts on the transcription rate of constitutive genes or on the promoter activation rate of telegraph genes through AND-combined Hill functions; per-gene rate constants can be drawn log-uniformly from ranges.

## Validation protocols

Exactness was tested by Kolmogorov-Smirnov distances between simulated stationary distributions and the analytic laws (Poisson for birth-death, Beta-Poisson for the telegraph model, negative binomial for bursty protein production<sup>46,47</sup>), by agreement between kernels, and by a two-sample comparison against JumpProcesses.jl<sup>48</sup> on the same model. Population-level properties were compared with the analytic expectations for cell-size scaling and concentration homeostasis<sup>23</sup>, partitioning noise<sup>49</sup>, lineage versus population statistics<sup>11</sup> and cell-cycle-dependent burst parameters<sup>10</sup>. Persister simulations were compared with the quantitative signatures reported by<sup>6,30</sup> and<sup>9</sup>, and with the phenotypic-selection model of<sup>29</sup>. All scripts, seeds and outputs are in the `paper/` directory of the repository.

## Identifiability and numerical robustness of the calibration

Because six parameters were fitted to six measured fractions, the calibration was examined in three ways. A conditional profile moves one parameter at a time over its full range on a log grid of eleven points, holding the others at their fitted values, and records the root-mean-square error of the fate fractions without the memory prior; parameters whose error stays within the binomial standard error of the measurement over a wide range are not determined by these data. A two-dimensional slice does the same over the memory of the resistant state and the fraction of cells in it, the two quantities that trade off most directly, with the death parameters held at their fitted values; a slice was used in preference to a profile with re-optimisation because, at this cost per evaluation, the inner search did not reliably recover the fitted solution and reported spuriously poor fits at short memory. Numerical robustness was checked by halving the integration step and comparing the simulated fractions with the seed-to-seed spread at the fitted parameters.

Because the lethality of cisplatin is enhanced during replication, the position of a cell in its division cycle when the drug arrives is a competing explanation for the dose-invariance of death timing and for the fraction of cells that survive without dividing. To test it, the same six parameters were refitted together with a seventh, the cycle-independent fraction  $\beta$  of the death hazard, with the peak of killing at the middle of the cycle ( $\varphi_0 = 0.5$ ,

582  $w = 0.15$ ), under the same objective, bounds, optimiser budget and seeds as the main fit. The two fits are  
 583 compared by the root-mean-square error of the fate fractions without the memory prior, by their predictions  
 584 at the held-out concentration, and by a conditional profile over  $\beta$  with the remaining parameters held at the  
 585 cycle-dependent fit. The same gate ( $\beta = 0.25$ ,  $\varphi_0 = 0.5$ ,  $w = 0.15$ ) was then applied to the persister simulations  
 586 of Fig. 4 and the melanoma simulations of Fig. 8a-e, everything else unchanged, to test whether their conclusions  
 587 depend on the hazard being cycle-blind (Supplementary Note 13). The window position is a modelling choice  
 588 rather than a claim about either agent: what is tested is robustness to a strong cell-cycle gate of any kind.

## 589 Statistics and reproducibility

590 No statistical method was used to predetermine sample size, and no experimental data were generated in this  
 591 study; sample sizes are the numbers of simulated cells, founders or seeds stated in each figure legend and in  
 592 Supplementary Table 2. No data were excluded. Randomisation and blinding do not apply: every result is  
 593 a deterministic function of an explicit random seed, and all seeds are fixed in the scripts that generate the  
 594 figures. Replicates are independent simulations differing only in seed; where a mean and a standard deviation  
 595 are reported, the number of seeds is given in the legend. Kolmogorov-Smirnov distances are compared with the  
 596 99% critical value for the stated sample size. Distributional agreement between kernels and with an independent  
 597 implementation was assessed by two-sample Kolmogorov-Smirnov statistics. Correlations are Pearson correlations  
 598 unless stated otherwise; correlations of binary fates between related cells are phi coefficients. Posterior summaries  
 599 are medians, because the posteriors are asymmetric on the log scale. Multithreaded runs give results identical to  
 600 single-threaded runs, because each cell carries its own random number generator seeded from its parent.

## 601 Data availability

602 The laboratory and clinical reference values used for calibration and validation were transcribed from the published  
 603 articles cited in Supplementary Table 4 and are provided, with their provenance, in the **paper/data/** directory  
 604 of the software repository (<https://github.com/ayoublasri/Biomodelling.jl>). All other data in this Article are  
 605 simulated and are regenerated from fixed seeds by the scripts in the **paper/** directory; the generated tables are  
 606 archived with the software release.

## 607 Code availability

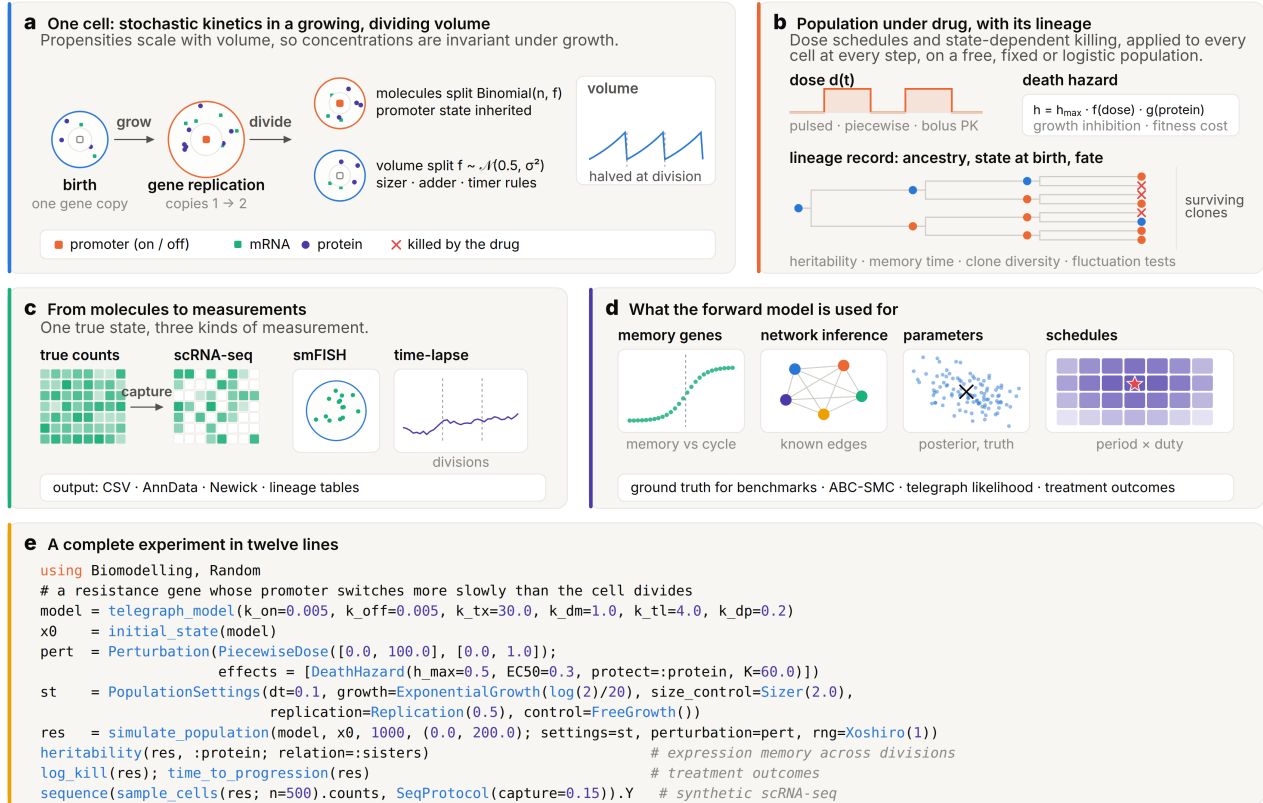
608 Biomodelling.jl 2.0 is open source under the MIT licence at <https://github.com/ayoublasri/Biomodelling.jl>, with  
 609 documentation, tutorials and a test suite that runs under continuous integration. Every figure is regenerated  
 610 from fixed seeds by the scripts in **paper/**. After instantiating the environment (`julia --project=paper -e`  
 611 `'using Pkg; Pkg.develop(path="."); Pkg.instantiate()'`), the simulation data are produced by `julia`  
 612 `--project=paper -t 4 paper/run_all.jl`, the Python network-inference and imputation benchmarks of Fig.  
 613 5b-d by `python paper/scripts/benchmarks_python.py`, and the identifiability analysis of Supplementary Fig.  
 614 3 by `julia --project=paper paper/scripts/fig7_calibration.jl profile`; the panels are then assembled  
 615 by `python paper/fig1_schematic.py`, `python paper/plot_all.py` and `python paper/plot_supplement.py`.  
 616 Full instructions, including the four-thread timings of Supplementary Table 3, are in **paper/README.md**.

## 617 Software

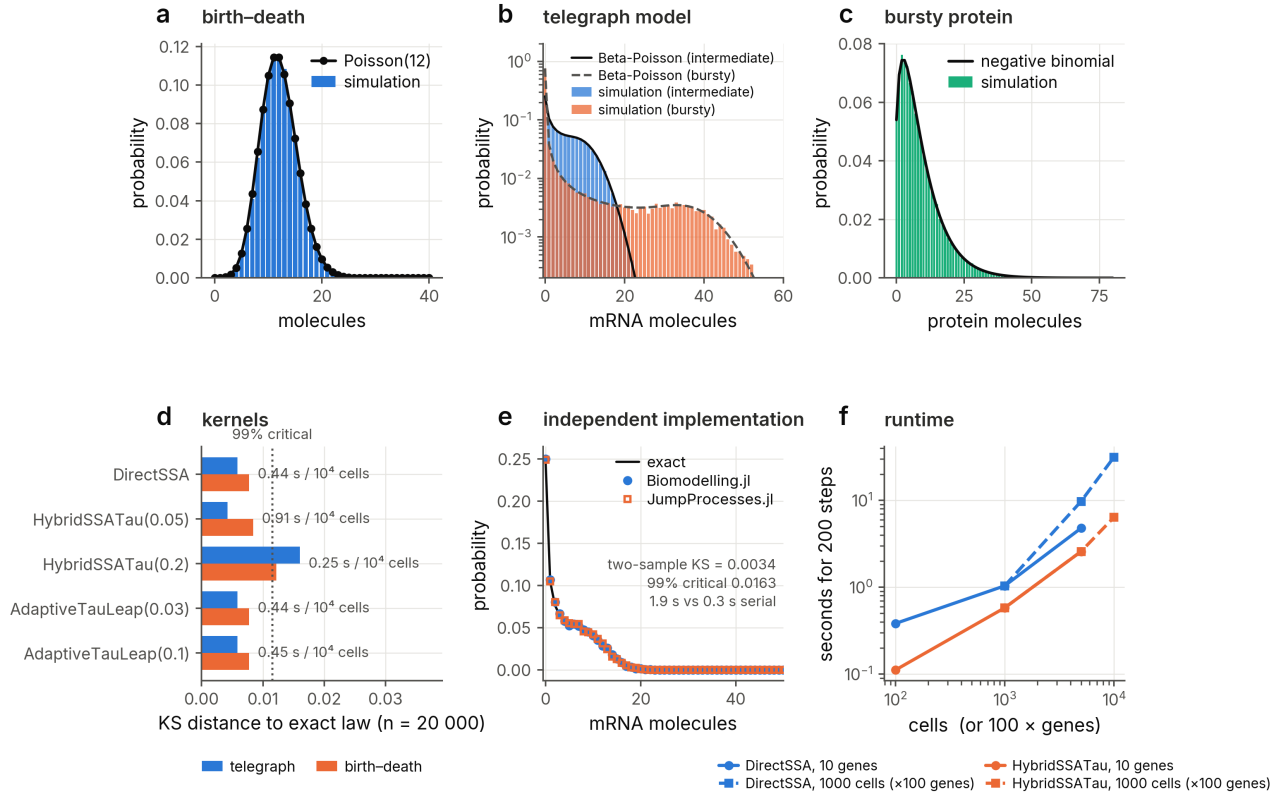
618 Biomodelling.jl 2.0 is written in Julia ( $\geq 1.10$ ) and depends on Distributions.jl, PoissonRandom.jl, QuadGK.jl,  
 619 Optim.jl, SpecialFunctions.jl and StatsBase.jl; HDF5.jl is an optional extension. Benchmarks used absolute  
 620 Pearson and Spearman correlations and scikit-learn random forests re-implementing GENIE3<sup>50</sup>, MAGIC<sup>51</sup> and

kNN-smoothing<sup>52</sup>. The test suite (Aqua quality checks and statistical tests) runs under GitHub Actions on Julia 1.10 and the latest release.

## Figure legends

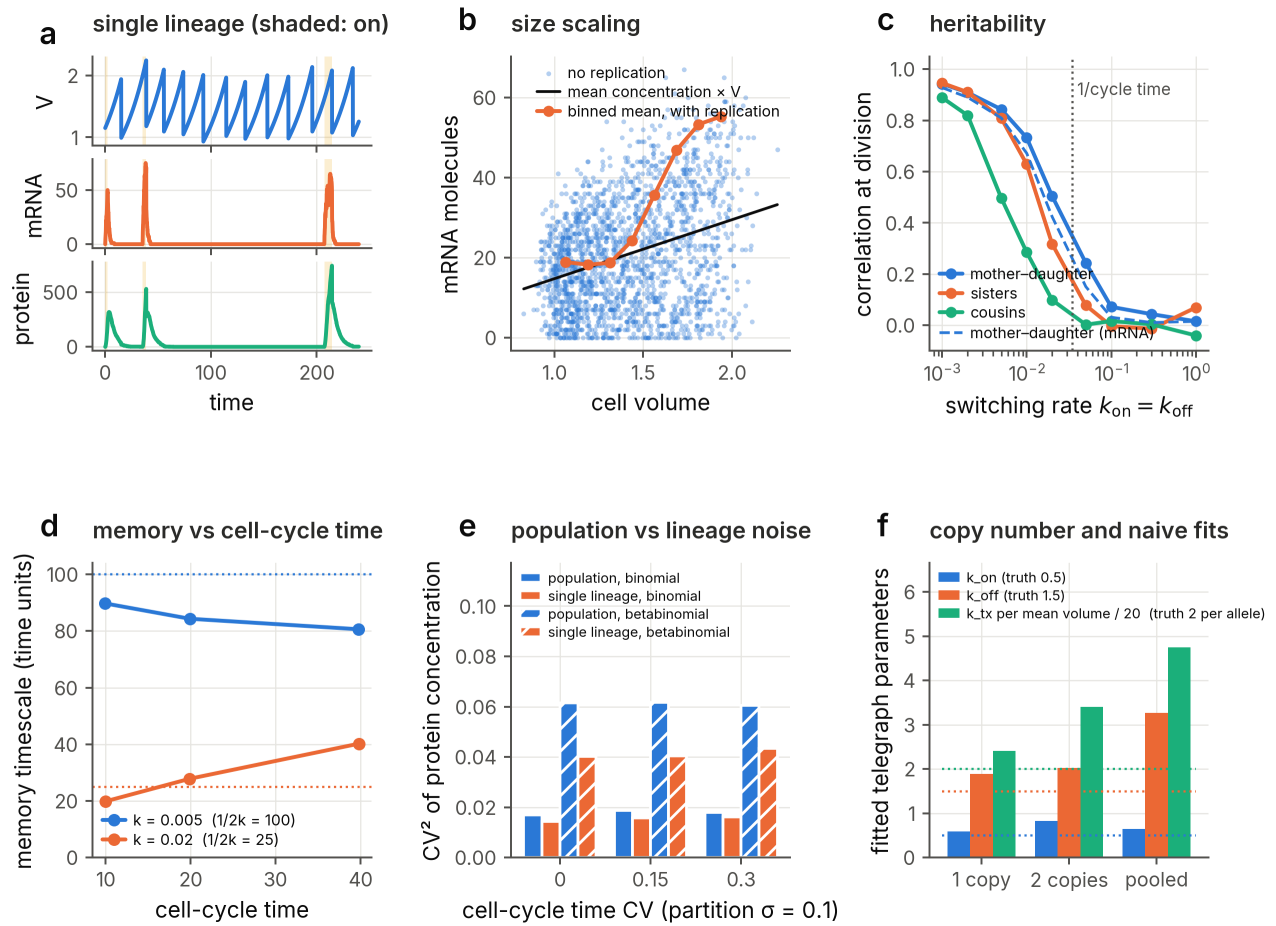


**Fig. 1 | The Biomodelling.jl 2.0 framework.** **a**, One cell. Reactions run at the cell's current volume, with propensities scaled so that concentrations are invariant under growth; genes replicate at a set point of the cycle; division splits every molecule binomially while both daughters inherit the promoter state. Inset: exponential volume growth, halved at each division. **b**, Populations under free growth, constant-size or logistic control, with dose schedules and a state-dependent death hazard evaluated for every cell at every step, and the lineage record from which heritability, autocorrelation, memory timescales, clone diversity and fluctuation tests are computed; crosses mark cells killed by the drug. **c**, Observation models turn true counts into scRNA-seq counts with capture, depth, dropout and batch effects, into smFISH spots, or into time-lapse reporter traces; outputs are CSV, AnnData and Newick. **d**, Uses of the forward model: detection of memory genes, benchmarking of network and perturbation inference against known ground truth, likelihood-free parameter inference, and schedule optimisation over the period and the fraction of time on drug (star, optimum). **e**, A complete experiment in twelve lines of code.



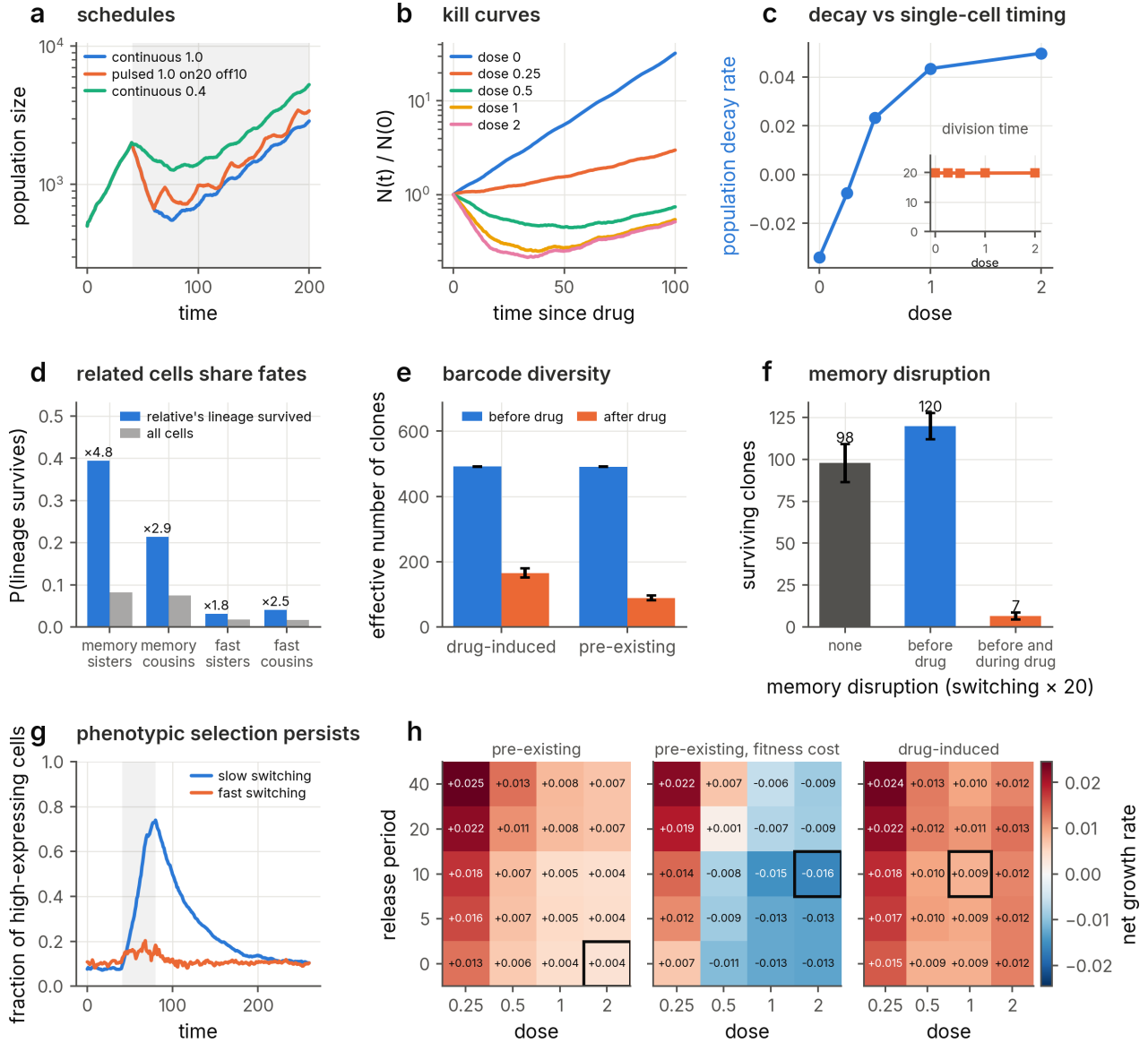
637

638 **Fig. 2 | The engine matches exact laws and an independent implementation.** **a**, Stationary distribution  
639 of a birth-death process (bars, 50,000 simulated cells; line, Poisson law). **b**, Telegraph model in an intermediate  
640 and a bursty regime against the Beta-Poisson law. **c**, Bursty protein production against the negative binomial. **d**,  
641 Kolmogorov-Smirnov distance to the exact law for every kernel (n = 20,000; dotted line, 99% critical value) with  
642 wall-clock time per  $10^4$  cells. **e**, Telegraph model simulated with Biomodelling.jl and with JumpProcesses.jl on  
643 the same model; the two-sample Kolmogorov-Smirnov statistic lies below its 99% critical value. **f**, Runtime of  
644 population simulations against the number of cells (10 genes) and genes (1,000 cells) for the exact and hybrid  
645 kernels.



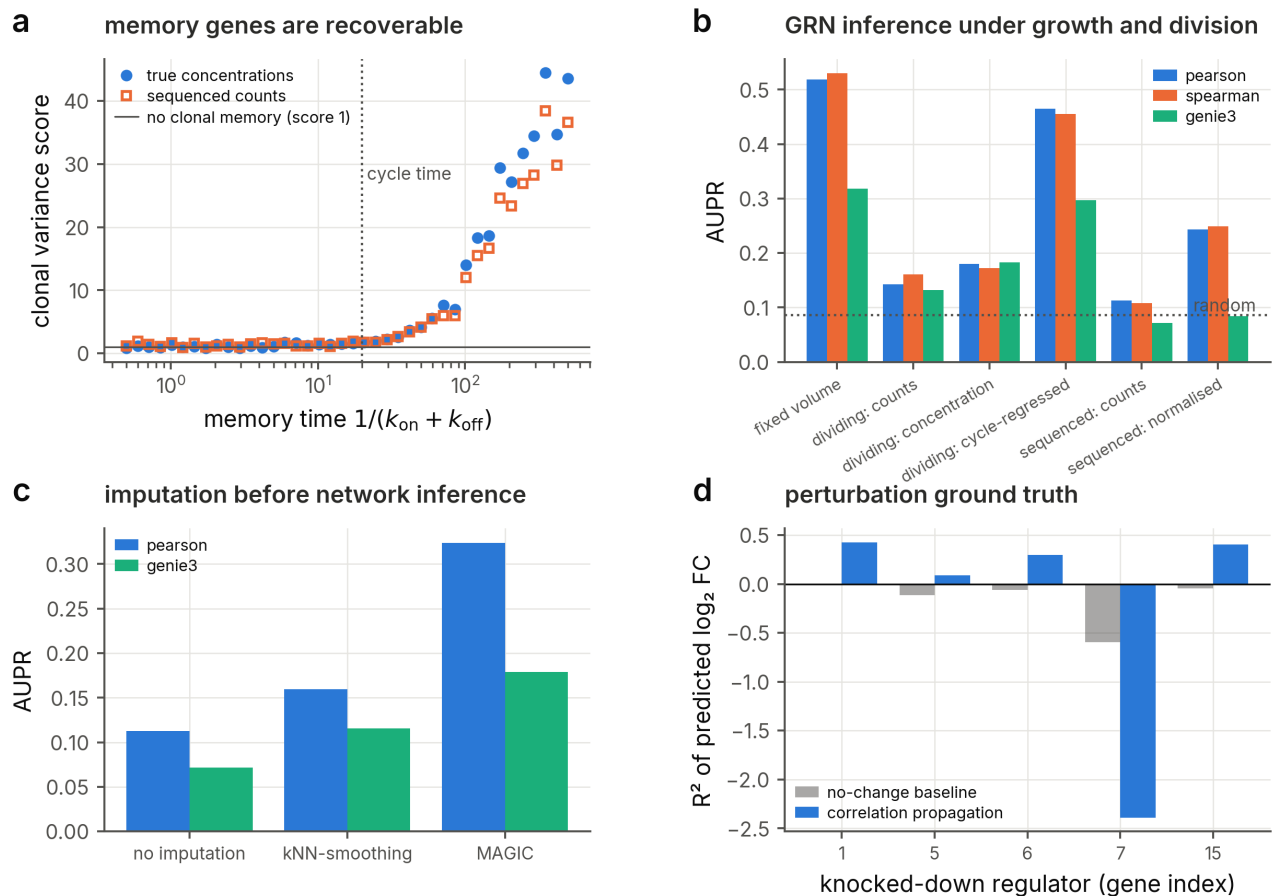
646

647 **Fig. 3 | Growth, division and partitioning generate memory, size scaling and cell-cycle effects.** **a**, A  
 648 single lineage: volume, mRNA and protein of a slowly switching gene, with the promoter-on state shaded. **b**,  
 649 mRNA count against cell volume for a fast-switching gene without replication (points; line, mean concentration  
 650  $\times V$ ) and with gene replication at mid-cycle (binned means). **c**, Correlation at division between mothers  
 651 and daughters, sisters and cousins as a function of the promoter switching rate (protein concentration, solid;  
 652 mother-daughter mRNA, dashed); dotted line, inverse cell-cycle time. **d**, Memory timescale against the measured  
 653 cell-cycle time for two switching rates; dotted lines,  $1/(k_{on} + k_{off})$ . **e**, Squared coefficient of variation of a stable  
 654 protein's concentration across the population and along single lineages, for increasing partitioning noise and  
 655 cell-cycle-time noise. **f**, Telegraph parameters fitted to cells before replication (one gene copy), after replication  
 656 (two copies) and pooled; dotted lines, per-allele values.



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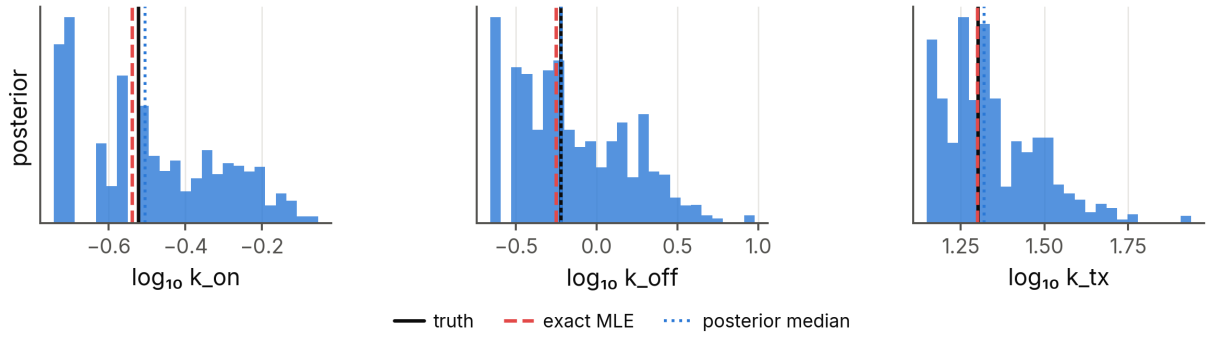
658 **Fig. 4 | A single resistance gene with slow promoter switching reproduces persister biology.** **a**,  
 659 Population size under continuous, pulsed and reduced-dose schedules (shading, treatment window). **b**, Kill curves  
 660 at increasing dose. **c**, Net population decay rate over the first 30 time units of treatment against dose, with  
 661 the mean division time of cells born under drug (inset), which is unchanged. **d**, Probability that a cell's lineage  
 662 survives the treatment (at least one descendant alive at the end of the run) given that its sister's or cousin's  
 663 lineage survived, against the marginal survival probability, for cells born in the 25 time units before the drug;  
 664 memory gene and fast-switching control, with the fold enrichment above each bar. **e**, Effective number of clones  
 665 (barcodes) before and after drug for pre-existing and drug-induced tolerance (mean  $\pm$  s.d. of three seeds). **f**,  
 666 Surviving clones after drug without memory disruption, with promoter switching accelerated 20-fold from two  
 667 cell-cycle times before the drug until the drug arrives, and with the acceleration maintained during exposure  
 668 (mean  $\pm$  s.d. of five seeds). **g**, Fraction of high-expressing cells before, during and after a drug pulse for slow  
 669 and fast switching. **h**, Net population growth rate over a 200-time-unit course as a function of release period  
 670 (time units without drug between 20-time-unit exposures) and dose, for pre-existing tolerance without and with a  
 671 fitness cost of the resistant state and for drug-induced tolerance; the box marks the most effective schedule.



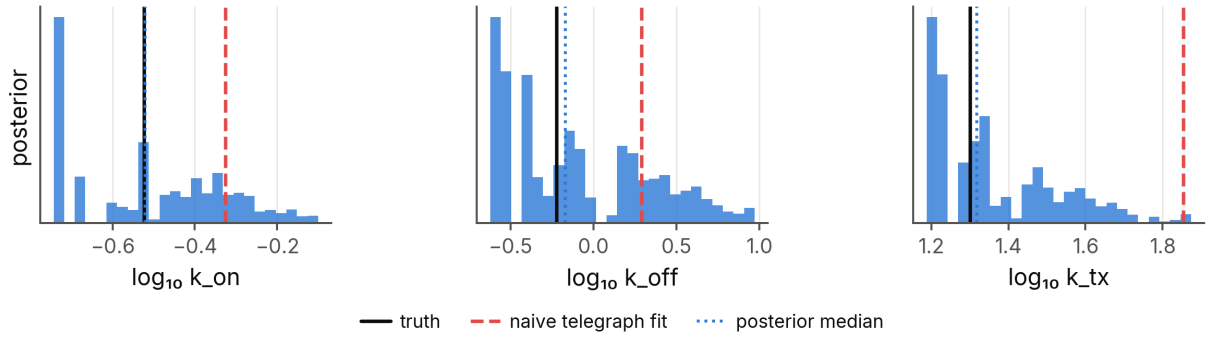
672

673 **Fig. 5 | Ground-truth benchmarks.** **a**, Clonal variance score against the known memory time of 40  
 674 independent genes, from true concentrations and from sequenced counts; dotted line, cell-cycle time. **b**, Area  
 675 under the precision-recall curve of correlation (Pearson, Spearman) and GENIE3 network inference on the same  
 676 30-gene network observed at fixed volume, in a dividing population (raw counts, concentrations, and concentrations  
 677 regressed on the cell-cycle covariates copy number and age) and after sequencing (raw and library-size normalised  
 678 counts); dotted line, random ranking. **c**, Effect of kNN-smoothing and MAGIC imputation before inference. **d**,  
 679 Coefficient of determination of predicted  $\log_2$  fold changes after knocking down the five most connected regulators,  
 680 for a no-change baseline and for propagation of observational correlations.

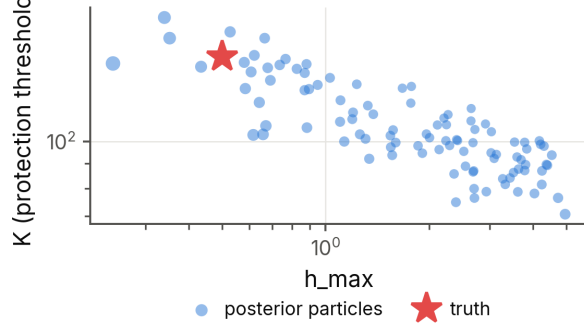
**a non-dividing cells: ABC vs exact MLE**



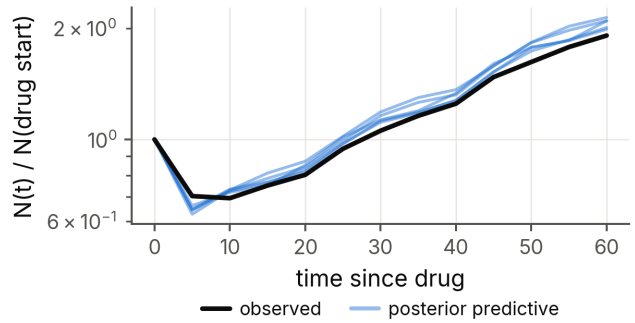
**b dividing population: division-aware ABC vs naive fit**



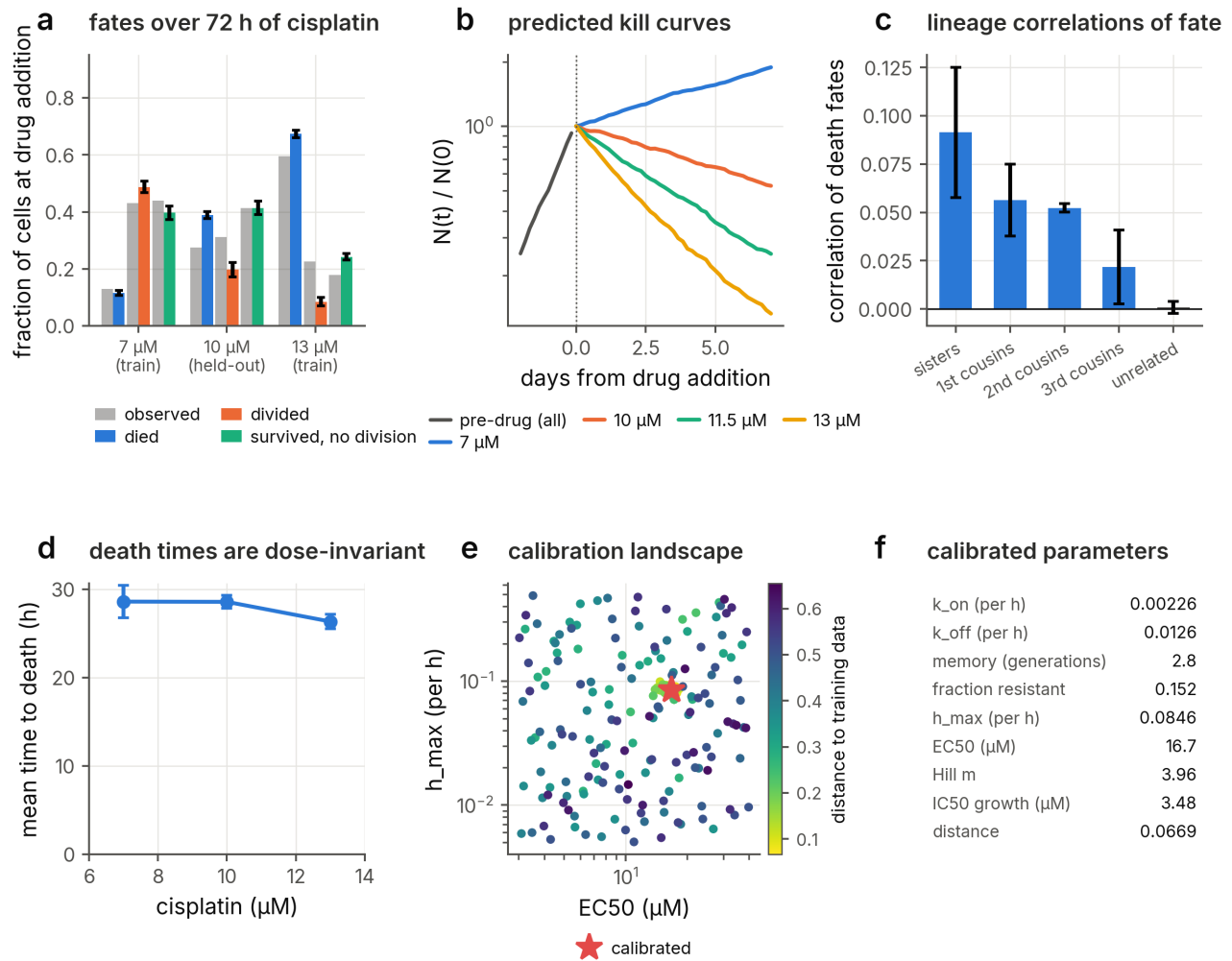
**c drug parameters from kill curve and sister fates**



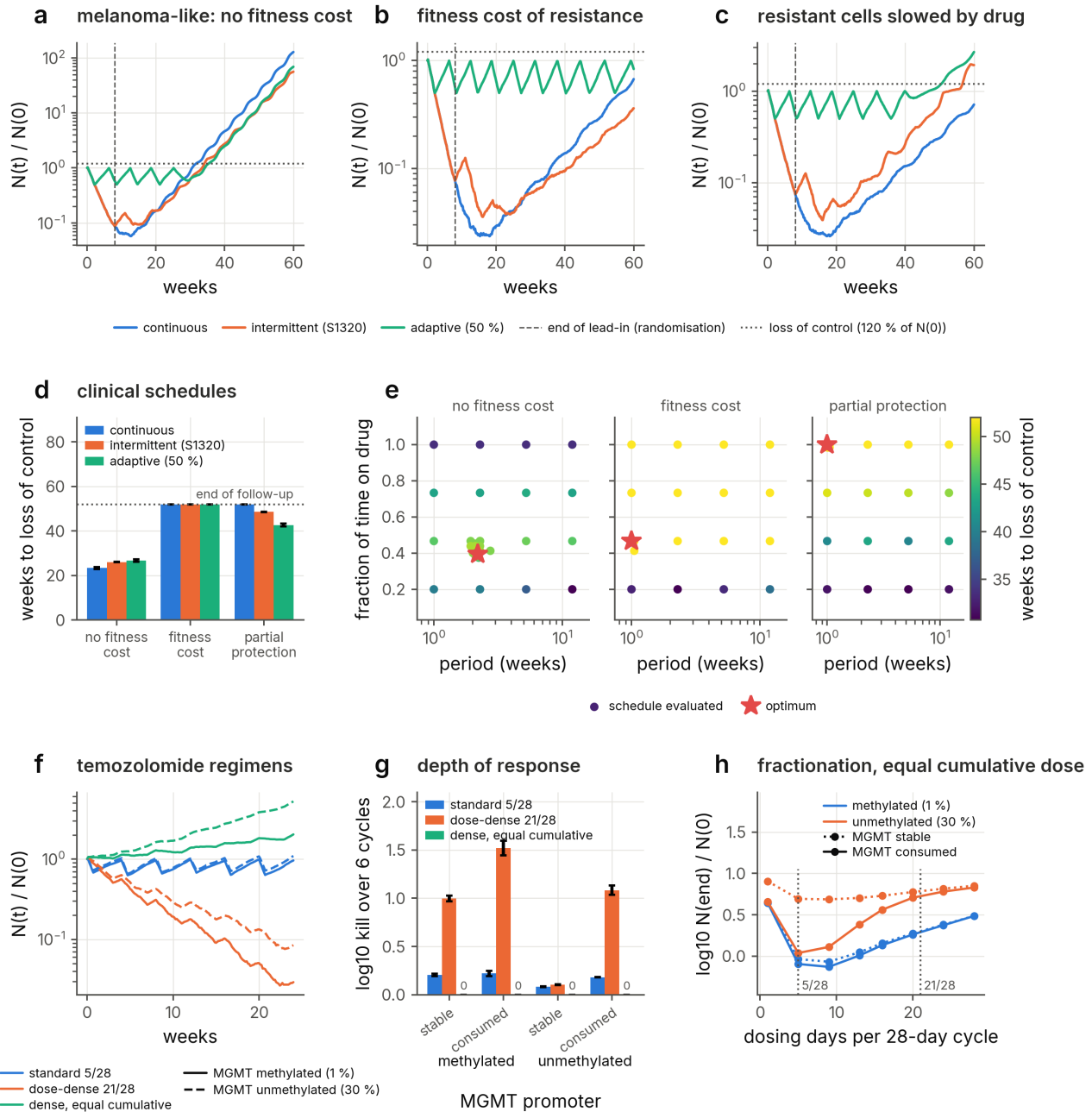
**d posterior predictive check**



**Fig. 6 | Inference with the forward model.** **a**, Posterior distributions of telegraph parameters from ABC-SMC on a snapshot of non-dividing cells, with the truth and the exact maximum-likelihood estimate. **b**, The same for a snapshot of a growing and dividing population: the naive telegraph fit is biased while division-aware ABC recovers the truth. **c**, Posterior particles for the maximal death rate and the protection threshold inferred from a kill curve and sister fate concordance: the two parameters trade off along a ridge that contains the truth. **d**, Posterior predictive kill curves simulated at the posterior median against the observed curve.



**Fig. 7 | Calibration to laboratory time-lapse data and validation on held-out conditions.** **a**, Fates over 72 h of cisplatin of U2OS cells present at drug addition, observed by time-lapse microscopy (grey; ref.<sup>6</sup>) and simulated by the calibrated model (colours; mean  $\pm$  s.d. of four seeds); the model was calibrated on 7 and 13  $\mu\text{M}$  and the 10  $\mu\text{M}$  condition is predicted. **b**, Predicted population size at the three concentrations and at 11.5  $\mu\text{M}$ , from two days before the drug to seven days of exposure. **c**, Correlation of death fates between sisters, first, second and third cousins present at drug addition, and between unrelated cells (mean  $\pm$  s.d. of three seeds), against the observation that correlations extend to second but not third cousins. **d**, Mean time to death of cells present at drug addition against concentration (mean  $\pm$  s.d. of four seeds); division times are not shown because too few cells born under drug divided again to estimate them (at most four per condition). **e**, Calibration landscape: every parameter set evaluated by the optimiser, coloured by its distance to the training data. **f**, Calibrated parameters.



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**Fig. 8 | Dose and schedule optimisation against clinical regimens.** **a–c**, Population size, relative to its pre-treatment value, under continuous dosing, the intermittent schedule of the SWOG S1320 trial (8-week lead-in, dashed line, then 3 weeks off and 5 weeks on) and adaptive therapy (drug withdrawn below 50% of the initial size, resumed at the initial size) for a melanoma-like population with a rare, heritable pre-resistant state, when the resistant state carries no fitness cost (**a**), when resistant cells grow 50% slower in the absence of drug (**b**), and when resistant cells are only partly protected and grow at half speed under drug (**c**); dotted line, 120% of the initial size. **d**, Weeks after randomisation until the population exceeds 120% of its pre-treatment size (loss of control; mean  $\pm$  s.d. of four seeds, censored at 52 weeks). **e**, The same end point over intermittent schedules of given period and fraction of time on drug after the lead-in, for every schedule evaluated by the optimiser (star, best), for the three mechanisms. **f**, Population size under the standard (5 of 28 days) and dose-dense (21 of 28

days) temozolomide regimens of the RTOG 0525 trial and a 21-day regimen at the standard cumulative dose, with one-compartment pharmacokinetics (elimination half-life 2.1 h) and stoichiometric consumption of MGMT by the drug, one molecule per lesion repaired, for populations in which 1% (methylated promoter, solid) or 30% (unmethylated, dashed) of cells express MGMT. **g**, Depth of response over six cycles for the three regimens, the two populations and MGMT either stable or consumed by the drug (mean  $\pm$  s.d. of two seeds); a bar at zero marks a regimen that achieves no net kill (the 21-day schedule at the standard cumulative dose). **h**, Final population size against the number of dosing days per 28-day cycle at the standard cumulative dose.

## Acknowledgements

Version 1 of the framework was developed with Marc Sturrock and Vahid Shahrezaei, whose contributions to that work are described in ref.<sup>25</sup>. This Article and version 2.0 of the software were authored with Claude (Anthropic), an AI assistant, under the direction and supervision of the author, who verified every result and takes full responsibility for the content.

## Author contributions

A.L. conceived the study, defined its scope, the validation strategy and the case studies, supervised the work, verified all results and wrote the final version of the manuscript. The implementation of the software, the simulations, the analyses, the figures and the first draft of the manuscript were produced with Claude (Anthropic) under the author’s direction.

## Competing interests

The author declares no competing interests.

## Additional information

**Supplementary information** accompanies this Article. Correspondence and requests for materials should be addressed to A.L.

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